

25.6 Regression adjustment in randomized experiments

In an experiment covariates are not needed to remove bias (Theorem 25.2.3); they are used to reduce variance, which analysis of covariance does under a normal linear model with a common slope (Proposition 18.1.2). Freedman (2008a) analysed adjustment in Neyman's framework of [Section 25.2](#), with no model, and found that the adjusted estimator is biased in finite samples, that its usual standard error can be wrong, and that it can be *less* precise than the difference in means. Lin (2013) showed that interacting the covariates with the treatment repairs the precision problem. This section proves both halves.

25.6.1 The setting

A completely randomized experiment assigns n_1 of n units to treatment and $n_0 = n - n_1$ to control. Each unit has potential outcomes $y_i(1), y_i(0)$ and a vector $\mathbf{x}_i \in \mathbb{R}^q$ of covariates measured before randomization, hence fixed numbers unaffected by treatment, which we centre: $\sum_i \mathbf{x}_i = \mathbf{0}$. Write

$$\begin{aligned} \mathbf{S}_{xx} &= \frac{1}{n-1} \sum_i \mathbf{x}_i \mathbf{x}_i^\top, & \mathbf{s}_t &= \frac{1}{n-1} \sum_i \mathbf{x}_i (y_i(t) - \bar{y}(t)), \\ \boldsymbol{\beta}_t &= \mathbf{S}_{xx}^{-1} \mathbf{s}_t & (t = 0, 1), \end{aligned}$$

assuming $\mathbf{S}_{xx}$ positive definite. The vector $\boldsymbol{\beta}_t$ is the least squares slope of the potential outcomes $y_i(t)$ on $\mathbf{x}_i$ over *all* n units, a finite-population projection coefficient in the sense of [Section 6.11](#). No linear model is assumed.

With $\bar{Y}_t$ and $\bar{\mathbf{x}}_t$ the arm means, three estimators are in use: the **difference in means** $\hat{\tau}_D = \bar{Y}_1 - \bar{Y}_0$; **Fisher's** $\hat{\tau}_F$, the coefficient of T in the least squares regression of Y on $1, T, \mathbf{x}$ (the analysis of covariance of [Section 18.1](#)); and **Lin's** $\hat{\tau}_L$, the coefficient of T in the regression on $1, T, \mathbf{x}, T\mathbf{x}$. All three belong to one family. For fixed vectors $\mathbf{b}_1, \mathbf{b}_0 \in \mathbb{R}^q$ put

$$\hat{\tau}(\mathbf{b}_1, \mathbf{b}_0) = (\bar{Y}_1 - \mathbf{b}_1^\top \bar{\mathbf{x}}_1) - (\bar{Y}_0 - \mathbf{b}_0^\top \bar{\mathbf{x}}_0). \quad (25.6.1)$$

Each arm mean is corrected for the chance imbalance of its covariate mean. Then $\hat{\tau}_D = \hat{\tau}(\mathbf{0}, \mathbf{0})$. By Theorem 18.1.1(b,c), the analysis of covariance estimates the slope by the pooled within-arm slope $\hat{\boldsymbol{\beta}}_p$ and the treatment effect by the difference of the arm means adjusted with it, so $\hat{\tau}_F = \hat{\tau}(\hat{\boldsymbol{\beta}}_p, \hat{\boldsymbol{\beta}}_p)$. The regression with the interaction fits a separate line in each arm, so by Exercise (C1) (whose argument is unchanged for a vector covariate) $\hat{\tau}_L = \hat{\tau}(\hat{\boldsymbol{\beta}}_1, \hat{\boldsymbol{\beta}}_0)$, with $\hat{\boldsymbol{\beta}}_t$ the least squares slope within arm t .

25.6.2 Fixed coefficients: an exact result

If the coefficients in [Eq. 25.6.1](#) are fixed in advance, everything is exact.

Lemma 25.6.1 — Adjustment with fixed coefficients

For fixed $\mathbf{b}_1, \mathbf{b}_0$, put $\mathbf{c} = (n_0\mathbf{b}_1 + n_1\mathbf{b}_0)/n$ and $\mathbf{c}^* = (n_0\boldsymbol{\beta}_1 + n_1\boldsymbol{\beta}_0)/n$. Then:

- (a) $E \hat{\tau}(\mathbf{b}_1, \mathbf{b}_0) = \tau$;
- (b) the variance is

$$\begin{aligned} \text{Var } \hat{\tau}(\mathbf{b}_1, \mathbf{b}_0) &= \text{Var } \hat{\tau}_D \\ &+ \frac{n}{n_0 n_1} \left[(\mathbf{c} - \mathbf{c}^*)^\top \mathbf{S}_{xx} (\mathbf{c} - \mathbf{c}^*) - \mathbf{c}^{*\top} \mathbf{S}_{xx} \mathbf{c}^* \right]. \end{aligned}$$

In particular the variance depends on $(\mathbf{b}_1, \mathbf{b}_0)$ only through $\mathbf{c}$, and it is smallest, over all fixed coefficients, exactly when $\mathbf{c} = \mathbf{c}^*$, for example when $(\mathbf{b}_1, \mathbf{b}_0) = (\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)$.

Proof. Define adjusted potential outcomes $\tilde{y}_i(t) = y_i(t) - \mathbf{b}_t^\top \mathbf{x}_i$. Because the treated units are exactly those whose $\tilde{y}_i(1)$ is observed, $\hat{\tau}(\mathbf{b}_1, \mathbf{b}_0)$ is the difference in means for the $\tilde{y}$'s, and their average effect is $\tau - (\mathbf{b}_1 - \mathbf{b}_0)^\top \bar{\mathbf{x}} = \tau$ since the covariates are centred. Part (a) is Theorem 25.2.3(a). By Eq. 25.2.4 and Lemma 25.2.2, the variance is $\{n_0/(nn_1)\} S_u^2$, where

$$\tilde{u}_i = \tilde{y}_i(1) + \frac{n_1}{n_0} \tilde{y}_i(0) = u_i - \frac{n}{n_0} \mathbf{c}^\top \mathbf{x}_i, \quad u_i = y_i(1) + \frac{n_1}{n_0} y_i(0).$$

Expanding,

$$S_u^2 = S_x^2 - 2(n/n_0) \mathbf{c}^\top \mathbf{s}_{xu} + (n/n_0)^2 \mathbf{c}^\top \mathbf{S}_{xx} \mathbf{c},$$

with

$$\mathbf{s}_{xu} = \mathbf{s}_1 + (n_1/n_0) \mathbf{s}_0 = \mathbf{S}_{xx}(\boldsymbol{\beta}_1 + (n_1/n_0) \boldsymbol{\beta}_0) = (n/n_0) \mathbf{S}_{xx} \mathbf{c}^*.$$

Completing the square,

$$S_u^2 = S_x^2 + \frac{n^2}{n_0^2} \left[(\mathbf{c} - \mathbf{c}^*)^\top \mathbf{S}_{xx} (\mathbf{c} - \mathbf{c}^*) - \mathbf{c}^{*\top} \mathbf{S}_{xx} \mathbf{c}^* \right].$$

Multiplying by $n_0/(nn_1)$, and noting that $\{n_0/(nn_1)\} S_u^2 = \text{Var } \hat{\tau}_D$, gives (b). The bracket is smallest iff $\mathbf{c} = \mathbf{c}^*$, because $\mathbf{S}_{xx}$ is positive definite. ■

The optimal $\mathbf{c}^*$ weights the treated slope by the *control* fraction and the control slope by the *treated* fraction: by Eq. 25.2.4, $\mathbf{c}^*$ is, up to the factor n_0/n , the slope of $u_i = y_i(1) + (n_1/n_0) y_i(0)$ on $\mathbf{x}_i$.

25.6.3 Estimated coefficients: Freedman's critique and Lin's repair

Estimated coefficients cost exact unbiasedness but, in large experiments, nothing else. Consider a sequence of finite populations and experiments indexed by n , with covariates centred in each, and assume:

(A1) $n_1/n \rightarrow p \in (0, 1)$;

(A2) the mean and the covariance matrix, over the n units, of $\mathbf{w}_i = (\mathbf{x}_i^\top, y_i(1), y_i(0))^\top$ converge

to finite limits, and the limit of $\mathbf{S}_{xx}$ is positive definite;
 (A3) $n^{-1} \sum_i \|\mathbf{w}_i - \bar{\mathbf{w}}\|^4$ is bounded.

A sequence of random vectors $\mathbf{g}_n$ is **bounded in probability** if for every $\epsilon > 0$ there is an M with $\Pr(\|\mathbf{g}_n\| > M) < \epsilon$ for all n . If $\mathbf{g}_n$ is bounded in probability and $\mathbf{h}_n \rightarrow \mathbf{0}$ in probability, then $\mathbf{h}_n^\top \mathbf{g}_n \rightarrow 0$ in probability.

Theorem 25.6.2 — Regression adjustment in a completely randomized experiment

Under (A1)–(A3), let $\mathbf{b}_n = (n_1 \boldsymbol{\beta}_1 + n_0 \boldsymbol{\beta}_0)/n$.

- (a) $\hat{\boldsymbol{\beta}}_t - \boldsymbol{\beta}_t \rightarrow \mathbf{0}$ for $t = 0, 1$, and $\hat{\boldsymbol{\beta}}_p - \mathbf{b}_n \rightarrow \mathbf{0}$, in probability.
- (b) $\sqrt{n} \{\hat{\tau}_L - \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)\} \rightarrow 0$ and $\sqrt{n} \{\hat{\tau}_F - \hat{\tau}(\mathbf{b}_n, \mathbf{b}_n)\} \rightarrow 0$ in probability. So, to order $n^{-1/2}$, Lin's estimator behaves like the optimal fixed-coefficient estimator and Fisher's like the fixed-coefficient estimator with the common coefficient $\mathbf{b}_n$.
- (c) The variances of these approximating estimators satisfy

$$\text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) \leq \text{Var } \hat{\tau}(\mathbf{b}, \mathbf{b}) \quad \text{for every } \mathbf{b},$$

in particular $\text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) \leq \text{Var } \hat{\tau}_D$; and

$$\text{Var } \hat{\tau}(\mathbf{b}_n, \mathbf{b}_n) - \text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) = \frac{(n_1 - n_0)^2}{nn_0n_1} (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0)^\top \mathbf{S}_{xx} (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0), \quad (25.6.2)$$

which is zero iff $n_1 = n_0$ or $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_0$. Fisher's approximating estimator has *larger* variance than the difference in means iff $\mathbf{b}_n^\top \mathbf{S}_{xx} \mathbf{b}_n > 2 \mathbf{b}_n^\top \mathbf{S}_{xx} \mathbf{c}^*$.

Proof.

- (a) Fix components j, l of $\mathbf{w}$, and let z_i be $w_{ij} - \bar{w}_j$ or $(w_{ij} - \bar{w}_j)(w_{il} - \bar{w}_l)$. The treated units are a simple random sample of size n_1 , so by Lemma 25.2.2 the mean of the z_i over the treated arm has mean equal to their mean over all units and variance at most S_z^2/n_1 , where $S_z^2 \leq (n-1)^{-1} \sum_i z_i^2 \leq (n-1)^{-1} \sum_i (1 + \|\mathbf{w}_i - \bar{\mathbf{w}}\|^4)$, which is bounded by (A3). By (A1) and Chebyshev's inequality, every such arm mean differs from the corresponding mean over all units by a quantity tending to zero in probability. The within-arm covariance matrix of $\mathbf{w}$ is $n_1/(n_1 - 1)$ times the arm mean of the products minus the product of the arm means, so it differs from the covariance matrix over all units by a quantity tending to zero in probability. In particular the arm's $\mathbf{x}$ -covariance matrix and its covariance vector of $\mathbf{x}$ with $y(1)$ are close to $\mathbf{S}_{xx}$ and $\mathbf{s}_1$. Since the limit of $\mathbf{S}_{xx}$ is positive definite and matrix inversion is continuous there, and $\boldsymbol{\beta}_1$ converges by (A2), $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1 \rightarrow \mathbf{0}$ in probability. The control arm is the same. The pooled slope is

$$\begin{aligned} \hat{\boldsymbol{\beta}}_p &= [(n_1 - 1)\hat{\mathbf{S}}_{xx,1} + (n_0 - 1)\hat{\mathbf{S}}_{xx,0}]^{-1} \\ &\quad \times [(n_1 - 1)\hat{\mathbf{s}}_1 + (n_0 - 1)\hat{\mathbf{s}}_0] \end{aligned}$$

in terms of the arm moments, which by the same argument is close to $[(n - 2)\mathbf{S}_{xx}]^{-1} \mathbf{S}_{xx} [(n_1 - 1)\boldsymbol{\beta}_1 + (n_0 - 1)\boldsymbol{\beta}_0]$, and this differs from $\mathbf{b}_n$ by $O(1/n)$.

(b) By [Eq. 25.6.1](#),

$$\hat{\tau}_L - \hat{\tau}(\beta_1, \beta_0) = -(\hat{\beta}_1 - \beta_1)^\top \bar{\mathbf{x}}_1 + (\hat{\beta}_0 - \beta_0)^\top \bar{\mathbf{x}}_0.$$

By Lemma 25.2.2, $E \bar{\mathbf{x}}_1 = \mathbf{0}$ and $E \|\bar{\mathbf{x}}_1\|^2 = (1/n_1 - 1/n) \text{tr } \mathbf{S}_{xx}$, so $n E \|\bar{\mathbf{x}}_1\|^2$ is bounded, and $\sqrt{n} \bar{\mathbf{x}}_1$ is bounded in probability by Markov's inequality; likewise $\sqrt{n} \bar{\mathbf{x}}_0$. With (a), both products tend to zero in probability. The argument for $\hat{\tau}_F$ is identical, with $\hat{\beta}_p - \mathbf{b}_n$ in both places.

(c) Apply [Lemma 25.6.1](#). The pair (β_1, β_0) has $\mathbf{c} = \mathbf{c}^*$, the minimum, and $(\mathbf{b}, \mathbf{b})$ has $\mathbf{c} = \mathbf{b}$. For $\mathbf{b} = \mathbf{b}_n$, $\mathbf{b}_n - \mathbf{c}^* = \{(n_1 - n_0)/n\}(\beta_1 - \beta_0)$, which gives [Eq. 25.6.2](#). Comparing with $\mathbf{b} = \mathbf{0}$, the variance is larger iff

$$(\mathbf{b}_n - \mathbf{c}^*)^\top \mathbf{S}_{xx} (\mathbf{b}_n - \mathbf{c}^*) > \mathbf{c}^{*\top} \mathbf{S}_{xx} \mathbf{c}^*,$$

which simplifies to the stated condition. ■

Part (b) concerns the leading term only. Because the slopes and $\bar{\mathbf{x}}_t$ come from the same assignment, $\hat{\tau}_L$ and $\hat{\tau}_F$ have a bias of order $1/n$ under moment conditions, negligible beside a standard error of order $n^{-1/2}$ (Freedman 2008a; Lin 2013). With a finite-population central limit theorem (Li and Ding 2017), part (b) makes both estimators asymptotically normal with the variances of part (c).

Freedman's critique is part (c) for Fisher's estimator: its common slope converges to $\mathbf{b}_n$, which weights each arm's slope by the arm's *own* size, while the optimal $\mathbf{c}^*$ uses the *other* arm's size. With an unbalanced design and an effect that varies with the covariate, the analysis of covariance can lose to no adjustment at all. **Lin's repair** is part (c) for the interacted regression, which converges to the optimal coefficients and is asymptotically at least as precise as both alternatives *whether or not the linear model is true*. When $n_1 = n_0$ the two adjusted estimators agree to first order.

Example. (When adjustment hurts)

A synthetic population of $n = 400$ units has one standardized covariate, on which the response depends under treatment ($\beta_1 = 2.007$) but not under control ($\beta_0 = -0.009$); $\tau = 1.003$. With $\beta_0 \approx 0$ and $S_{xx} = 1$, the condition of part (c) reads $p\beta_1(p\beta_1 - 2(1-p)\beta_1) > 0$, that is $p > 2/3$. [Figure 25.6.1](#) compares n times the variance of each estimator over 20000 randomizations with the exact variances of the approximating fixed-coefficient estimators:

fraction treated	difference in means	Fisher (ANCOVA)	Lin (interacted)
0.2	20.88 (20.81)	14.03 (13.87)	4.80 (4.73)
0.5	6.01 (6.02)	2.04 (2.03)	2.03 (2.03)
0.8	5.20 (5.20)	13.50 (13.37)	4.26 (4.23)

(simulated; values from [Lemma 25.6.1](#) in brackets). At $p = 0.8$ the analysis of covariance has more than twice the variance of the unadjusted estimate while Lin's estimator still gains; at $p = 0.5$ both cut it by two thirds.

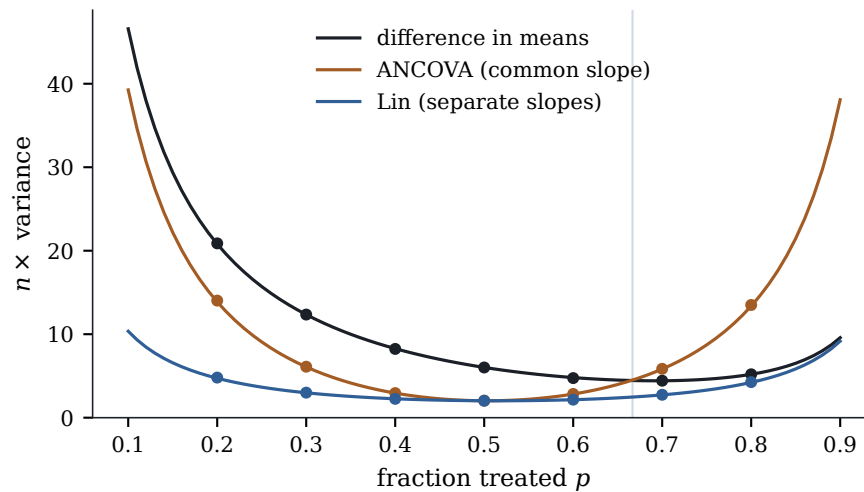


Figure 25.6.1: n times the variance of the three estimators in [Example \(When adjustment hurts\)](#) as a function of the fraction treated. Curves: exact variances of the fixed-coefficient estimators that the estimators approach ([Theorem 25.6.2](#)). Points: simulation over 20000 randomizations. The vertical line is $p = 2/3$, beyond which the analysis of covariance loses to the difference in means.

```
import numpy as np

rng = np.random.default_rng(2506)
n = 400
x = rng.normal(size=n)
x = (x - x.mean()) / x.std(ddof=1)           # centred,  $S_{xx} = 1$ 
y0 = rng.normal(size=n)                     # control: no dependence on  $x$ 
y1 = 1 + 2 * x + rng.normal(size=n)         # treated: slope 2
tau = np.mean(y1 - y0)
beta1, beta0 = np.polyfit(x, y1, 1)[0], np.polyfit(x, y0, 1)[0]

def estimators(T):
    """Difference in means, ANCOVA and Lin's estimator for assignment rows  $T$  (reps  $x$ 
     $n$ )."""
    n1 = T.sum(axis=1, keepdims=True); n0 = n - n1
    yobs = np.where(T, y1, y0)
    xb1, xb0 = (T * x).sum(1, keepdims=True) / n1, (~T * x).sum(1, keepdims=True) / n0
    yb1, yb0 = (T * yobs).sum(1, keepdims=True) / n1, (~T * yobs).sum(1, keepdims=True) / n0
    dx = np.where(T, x - xb1, x - xb0)           # deviations from the arm means
    dy = np.where(T, yobs - yb1, yobs - yb0)
    b1 = (T * dx * dy).sum(1) / (T * dx ** 2).sum(1) # within-arm slopes
    b0 = (~T * dx * dy).sum(1) / (~T * dx ** 2).sum(1)
    bp = (dx * dy).sum(1) / (dx ** 2).sum(1)        # pooled within-arm slope
    xb1, xb0, yb1, yb0 = xb1[:, 0], xb0[:, 0], yb1[:, 0], yb0[:, 0]
    unadj = yb1 - yb0
    ancova = unadj - bp * (xb1 - xb0)
    lin = (yb1 - b1 * xb1) - (yb0 - b0 * xb0)        #  $x$  is centred, so  $\bar{x} = 0$ 
    return unadj, ancova, lin
```

[Run this code online](#)

```

p, reps = 0.8, 2000
T = np.array([rng.permutation(n) < p * n for _ in range(reps)])
for name, est in zip(["difference in means", "ANCOVA", "Lin (interacted)"],
    estimators(T)):
    print(f"{name:20s} mean {est.mean():6.3f} (tau = {tau:.3f}) n * variance {n *
        est.var():6.2f}")

```

[Run this code online](#)

25.6.4 Standard errors

Freedman (2008a) showed that the classical standard error of the analysis of covariance can be inconsistent in either direction. Lin (2013) showed that the sandwich standard errors of Chapter 21 (Theorem 21.4.2) for the interacted regression are consistent or asymptotically conservative, for the reason seen in Exercise (B2): the residuals estimate the variances of Exercise (B1) arm by arm, and the one term that cannot be estimated enters with a negative sign.

In Example (When adjustment hurts) at $p = 0.8$, over 10000 randomizations, nominal 95% intervals covered τ with frequency 0.997 for the difference in means with Neyman's standard error, 0.921 for the analysis of covariance with its classical standard error, and 0.984 for Lin's estimator with the HC2 standard error. The first and last are conservative because the effect varies across units.

Idea. In a randomized experiment, adjust only for covariates measured before randomization; include their interactions with the treatment (with the covariates centred at their overall mean); and use sandwich standard errors. Then adjustment cannot hurt in large samples, whatever the true relation between covariates and responses, and it helps as much as a linear adjustment can.

Freedman (2008b) extends the critique to several treatments; the interacted regression, with one set of slopes per arm, extends Lin's repair. With many covariates relative to n the $O(1/n)$ terms are no longer negligible.

25.6.5 Exercises

25.6.5.1 A. Check your understanding

Exercise A1.

With one covariate and a fixed common coefficient b , show from Lemma 25.6.1 that $\hat{\tau}(b, b)$ has smaller variance than the difference in means iff b lies strictly between 0 and $2c^*$. What does this say about guessing a slope from a previous study?

Solution. With $\mathbf{c} = b$ and $S_{xx} > 0$, the variance difference is $\{n/(n_0n_1)\}S_{xx}[(b - c^*)^2 - c^{*2}] = \{n/(n_0n_1)\}S_{xx}b(b - 2c^*)$, which is negative iff b is strictly between 0 and $2c^*$. A slope guessed from earlier data, fixed before randomization, keeps the estimator exactly unbiased, and it reduces the variance as long as it has the right sign and is less than twice the optimal value: a crude guess is enough.

25.6.5.2 B. Practice

Exercise B1.

Let $e_i(t) = y_i(t) - \bar{y}(t) - \boldsymbol{\beta}_t^\top \mathbf{x}_i$ be the residuals of the finite-population regressions. Show that

$$\text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) = \frac{S_{e(1)}^2}{n_1} + \frac{S_{e(0)}^2}{n_0} - \frac{S_{e(1)-e(0)}^2}{n},$$

Neyman's formula with the potential outcomes replaced by their residuals. Deduce that the proportional gain over the difference in means is large when the covariates explain much of both potential outcomes.

Solution. As in the proof of [Lemma 25.6.1](#), $\hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)$ is the difference in means for the adjusted outcomes $y_i(t) - \boldsymbol{\beta}_t^\top \mathbf{x}_i = \bar{y}(t) + e_i(t)$. Adding constants changes nothing in Eq. 25.2.3, so its variance is Neyman's formula for the $e_i(t)$, whose unit effects are $e_i(1) - e_i(0)$ up to a constant. The first two terms are $(1 - R_t^2)$ times those of the unadjusted formula, where R_t^2 is the finite-population R^2 of $y(t)$ on $\mathbf{x}$.

Exercise B2.

In the setting of [Example \(When adjustment hurts\)](#), let $\beta_0 = 0$, $\beta_1 = \beta \neq 0$ and $S_{xx} = 1$, and write $p = n_1/n$. Show that the excess [Eq. 25.6.2](#), multiplied by n , is $(2p - 1)^2 \beta^2 / \{p(1 - p)\}$, and that the analysis of covariance loses to the difference in means iff $p > 2/3$. What happens if instead $\beta_1 = 0$ and $\beta_0 = \beta$?

25.6.5.3 C. Going deeper

Exercise C1.

Show that Lin's estimator is the average over *all* n units of the difference between the two arms' fitted regression lines: if $\hat{\alpha}_t + \hat{\boldsymbol{\beta}}_t^\top \mathbf{x}$ is the least squares fit in arm t , then

$$\hat{\tau}_L = n^{-1} \sum_i \{(\hat{\alpha}_1 + \hat{\boldsymbol{\beta}}_1^\top \mathbf{x}_i) - (\hat{\alpha}_0 + \hat{\boldsymbol{\beta}}_0^\top \mathbf{x}_i)\}.$$

Relate each arm's term to the regression estimator of a population mean in survey sampling.